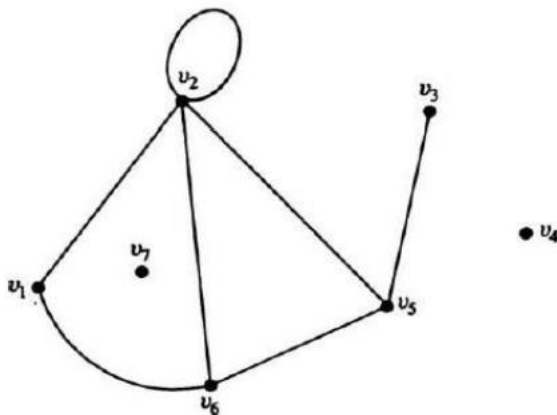
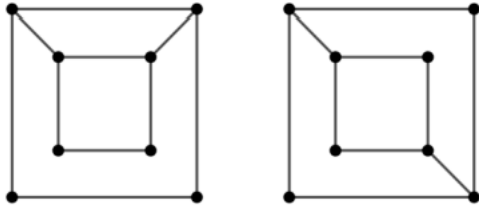


SHRI LABHUBHAI TRIVEDI INSTITUTE OF ENGINEERING & TECHNOLOGY, RAJKOT
SUBJECT: DISCRETE MATHEMATICS(3140708)
TUTORIAL-1

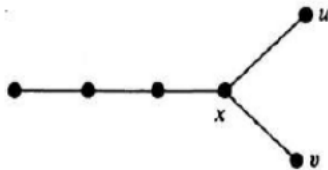
1. Define simple graph, degree of a vertex and complete graph. Illustrate each with an example
2. Define indegree and outdegree of a graph with example.
3. Draw a graph which is regular but not bipartite.
4. Define Cyclic graph, Null graph and Strongly connected graph.
5. Explain Path and Circuit of a graph. (or Define cycle, walk or Define path and circuit of a graph by drawing the graphs.)
6. How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2?
7. Prove that any graph has even number of odd vertices.
8. Define Isolated vertex, Pendent vertex and Size of a graph.
9. Does there exist a 4-regular graph with 6 vertices? If so, construct the graphs.
10. A graph G has 15 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G .
11. Define self-loop, adjacent vertices.
12. (i) Find the number of edges in G if it has 5 vertices each of degree 2.
(ii) Define complement of a subgraph by drawing the graphs.
13. Explain the following terms with proper illustrations. a) Directed graphs b) elementary path
c) Reachability of a vertex d) Connected graph.
14. Define null graph, centre of a graph and complete graph.
15. Define the terms: Multi – Graph, Weighted Graph. Illustrate each with an example.
16. Find the number of edges in a r -regular graph with n vertices.
17. 1) Define: Isolated vertex
2) Identify Isolated vertex/vertices from the following graph



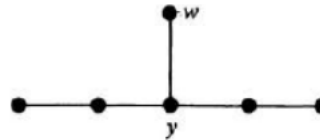
18. Let G be a graph with n vertices and m edges such that vertices have degree k or $k+1$. Prove that if G has N_k vertices of degree k and N_{k+1} vertices of degree $k+1$ then $N_k = (k+1)n - 2m$.
19. Define Isomorphic Graphs. Verify the following graphs are Isomorphic or not (Justify).



20. Verify the following graphs are Isomorphic or not (Justify).

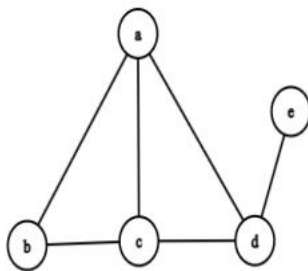


Graph -1

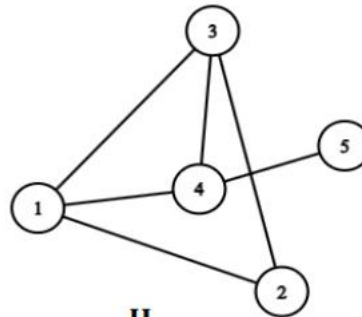


Graph-2

21. Determine whether the following graphs are isomorphic or not.

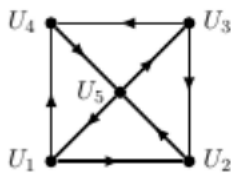


G

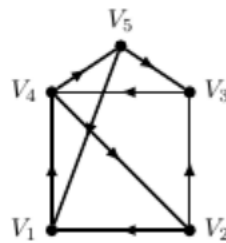


H

22. Check whether the following graphs are isomorphic or not.

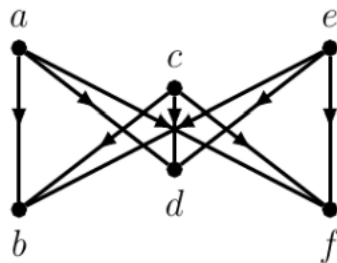


$G(V, E)$

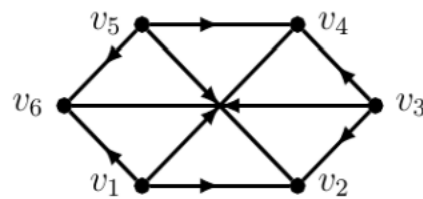


$H(V, E)$

23. Check whether the following graphs are isomorphic.

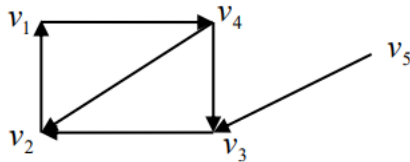


(H)

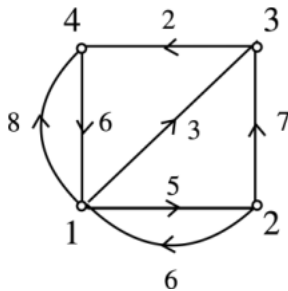


(G)

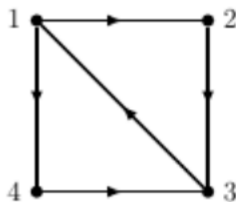
24. (i) A graph G has 15 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G . (ii) Define vertex disjoint and edge disjoint subgraphs by drawing the relevant graphs.
25. Define the following by drawing graphs (i) weak component (ii) unilateral component (iii) strong component.
26. Use Warshall's algorithm to obtain path matrix from the adjacency matrix of



27. Find the shortest path between each pair of vertices for a simple digraph using Warshall's Algorithm.



28. Find transitive closure by Warshall's Algorithm if $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1, 2), (3, 4), (4, 5), (4, 1), (1, 1)\}$
29. Define path matrix. Warshall's algorithm to obtain path matrix from the adjacency matrix of following graph



30. Define Eulerian graph, Planar Graph. Justify whether the graph given in Figure 1 is Planar or not using Euler's formula.

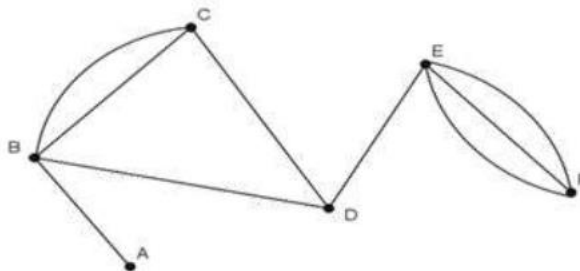
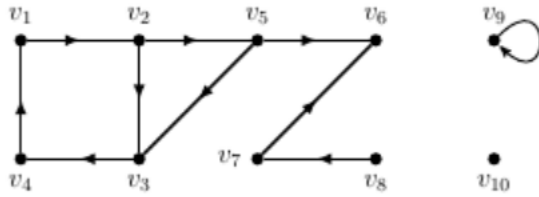
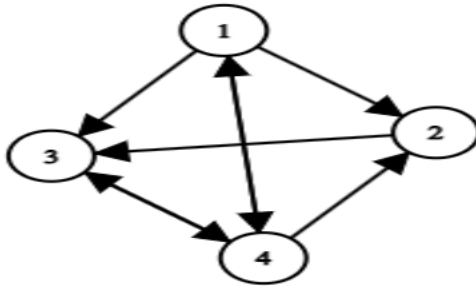


Figure 1

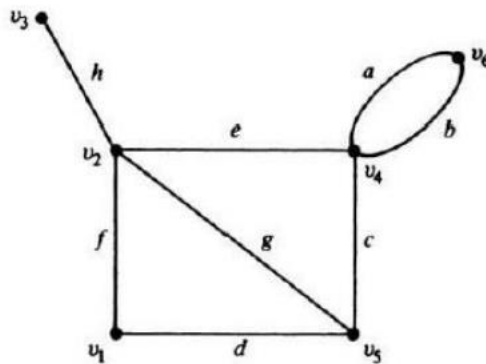
31. Find all the node base of the given digraph. Also find $d(V_3, V_6)$, $d(V_6, V_3)$.



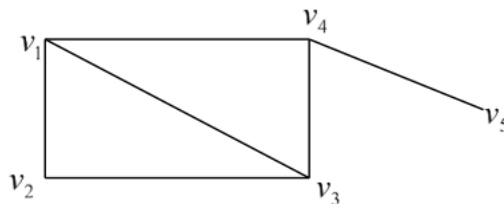
32. Define tree. Prove that there is one and only one path between every pair of vertices in a tree T.
33. (i) Find the number of edges in a r -regular graph with n -vertices. (ii) A tree T has 4 vertices of degree 2, 4 vertices of degree 3, 2 vertices of degree 4. Find the number of pendant vertices in T.
34. Find reachable set of all the vertices and node base of following graph.



35. 1) Define incidence Matrix of a Graph
2) Find incidence matrix for the following graph



36. Define adjacency matrix and find the same for



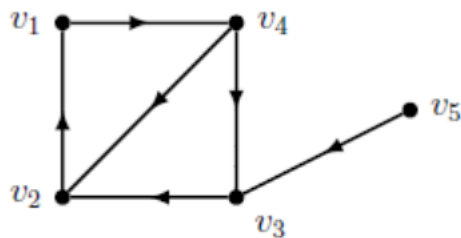
37. Let $S = \{1, 2, 3, 4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm.

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

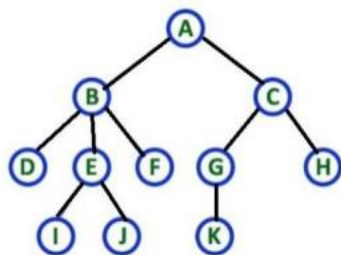
38. Define matrix of a relation. Let $A = \{a, b, c, d\}$ and let R be the relation on A that has the matrix

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

39. Define adjacency matrix. Use Warshall's algorithm to obtain path matrix from the adjacency matrix of

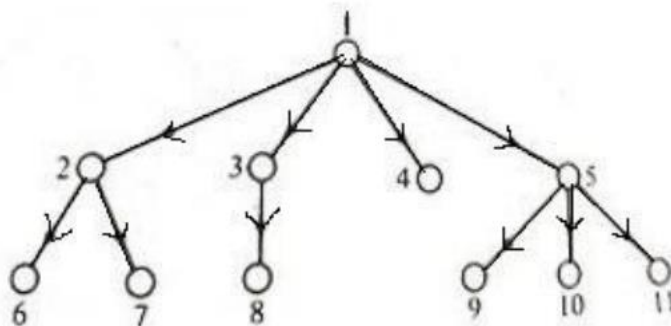


40. 1) Define Subtree and Degree of a Node
2) Determine degree of the each node for the following tree.



41. For the following set of weights construct an optimal binary prefix code. For each weight in the set, give corresponding code word. 5, 7, 8, 15, 35, 40

42. 1) Define: M-ary Tree and Binary Tree.
2) Represent the following directed tree as Binary tree



43. Prove that a tree with n vertices has $n - 1$ edges.
44. Define Isolated node, Binary tree and Regular graph.
45. Draw a binary tree whose post-order produced the string d g e b i j h f c a - - - - - and pre-order produces the string j b a c d i h e g f

46. Draw binary trees whose post-order produced the string d-e-c-g-j-h-f-b-l-n-q-r-p-m-k-a and pre-order produced the string a-b-d-h-e-i-j-cf-g-k and in-order produced the string h-d-b-i-e-j-a-f-c-k-g.
47. Prove that if a graph G has one and only one path between every pair of vertices then G is a tree.
48. (i) Does a 3-regular graph with 5 vertices exist?
(ii) Define centre of a graph and radius of a tree.
49. Show that sum of in-degrees of all the nodes of simple digraph is equal to the sum of out-degrees of all the nodes and this sum equal to the number of edges in it.
50. Define tree and root. Also prove that tree with n vertices has $n - 1$ edges.
51. Define adjacency matrix and path matrix of a graph. Find out adjacency matrix for the graph given in Figure 1.
52. Define Binary tree, Spanning tree, Minimal Spanning tree, Find the minimal spanning tree for the weighted graph given in Figure 2.

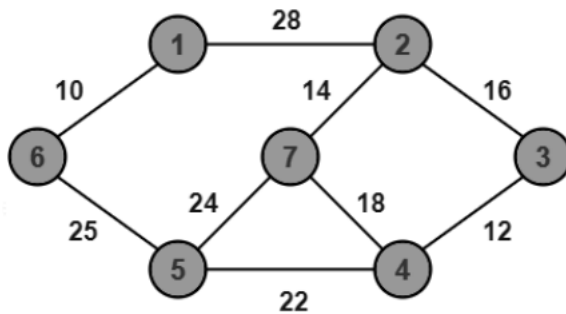


Figure 2

53. Form a binary search tree for the data 16,24,7,5,8,20,40,3.
54. Explain Post order traversal. Given the postorder and inorder traversal of a binary tree, draw the unique binary tree. Postorder: d e c f b h i g a Inorder: d c e b f a h g i.

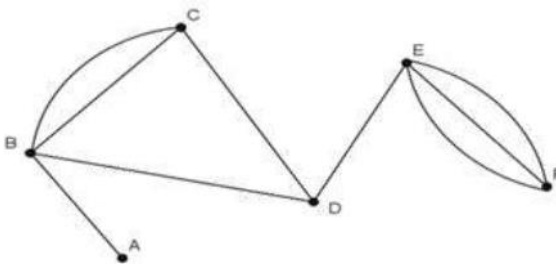


Figure 1

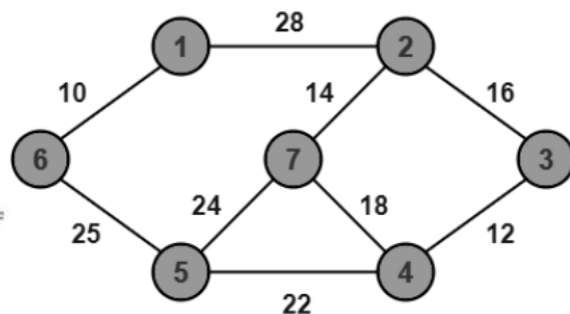
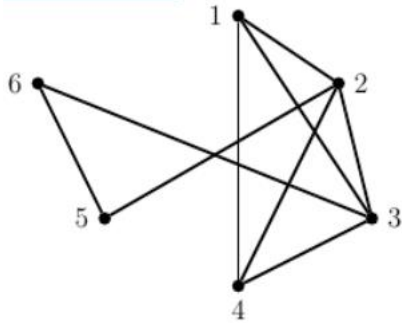


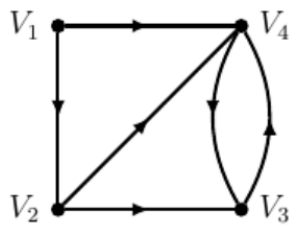
Figure 2

55. Find out maximal compatibility blocks of following simplified graph of digraph and write its relation matrix.

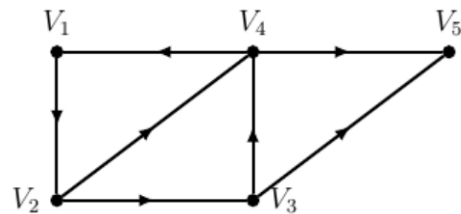


A tree T has 3 vertices of degree 4, 3 vertices of degree 3. Find the number of pendant vertices in tree T .

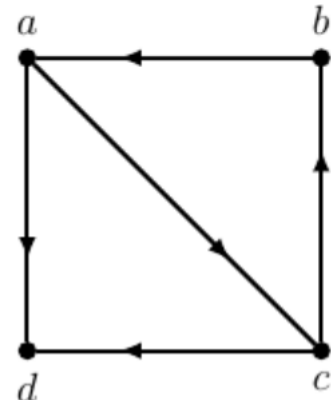
56. Find reachable set of each node of the given digraph. Also find $d(V_1, V_3)$, $d(V_3, V_1)$.



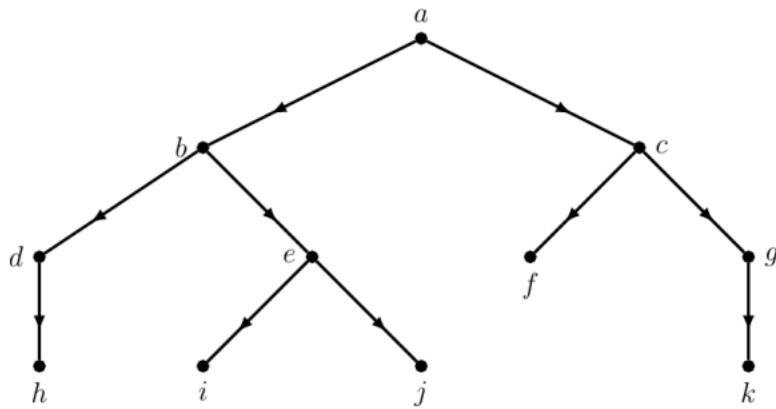
57. Define Strong, unilateral, weak component. Also Find strong, unilateral, weak component from the given digraph.



58. Find the path matrix using adjacency matrix of the following graph.

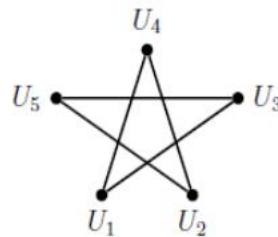
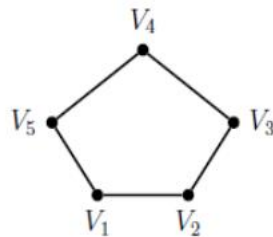


59. In which order does a pre-order, in-order and post-order traversal visit the vertices of the ordered rooted tree shown in figure?

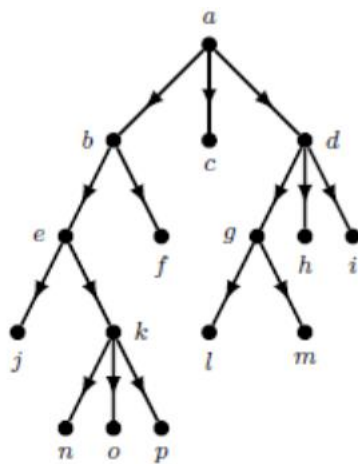


60. Define elementary cycle, loop, tree and pendent vertex.

61. Check whether the following graphs are isomorphic or not



62. In which order does a pre-order, in-order and post-order traversal visit the vertices of the ordered rooted tree shown in figure?



63. A tree T has 4 vertices of degree 2, 3 vertices of degree 3, 1 vertex of degree 4. Find the number of pendant vertices in the tree.